

4COSCo02W Mathematics for Computing

Lecture 11

Maths-Related Applications for Statistical Analysis

UNIVERSITY OF
WESTMINSTER

The Mathematics of Regression Analysis

Purpose of the Regression: Modeling the relationship between a dependent variable Y and one or more independent variables X_i .

Simple Linear Regression Formula: $Y = \beta_0 + \beta_1 X + \epsilon$

- Y : Dependent variable
- X : Independent variable
- β_0 : Intercept
- β_1 : Slope coefficient (rate of change in Y with X)
- ϵ : Error term (difference between observed and predicted values)

Multiple Linear Regression Formula: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$

Extends to multiple independent variables $X_1, X_2, \dots, X_n$

Key Metrics in Regression Analysis

Root Mean Squared Error (RMSE):

- **What It Measures:** Average error in model predictions.
- **Units:** Same as the dependent variable (e.g., litres).
- **Interpretation:**
 - Lower RMSE = Better model performance.
 - Example: RMSE = 10 means, on average, the model's predictions are 10 units off.

p-Value:

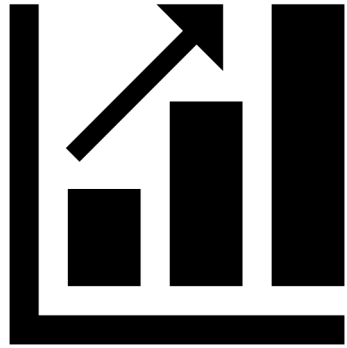
- **What It Measures:** Significance of the relationship between a predictor and the outcome.
- **Threshold:** p-value < 0.05 means the relationship is statistically significant.
- **Interpretation:**
 - Small p-value = Strong evidence of a real effect.
 - Large p-value = Weak or no evidence of an effect.

R-Squared (R^2):

- **What It Measures:** Proportion of variance in the outcome explained by the predictors.
- **Range:** 0 to 1 (or 0% to 100%).
- **Interpretation:**
 - $R^2=0.8$: Model explains 80% of the variability in the outcome.
 - Higher R^2 = Better model fit.

Key Metrics in Regression Analysis

Metric	Purpose	Ideal Value
RMSE	Measures average prediction error	Smaller is better
p-Value	Tests significance of variable relationships	Less than 0.05
R-Squared (R^2)	Measures how well the model explains the data	Closer to 1 is better



Statistical Analysis Applications

WEKA, EViews, MATLAB

Apps Demonstration

Using the Sample Dataset: Petrol Consumption

We will analyse the **Petrol Consumption Dataset** (MathCompW11-SampleDataset-PetrolConsumption.xls) provided on Blackboard using **WEKA**, **EViews**, and **MATLAB** to perform multiple linear regression.

Objectives:

1. Set up and execute a regression model in each tool.
2. Interpret outputs and compare model performance across tools.

Dataset Overview: Review variables (e.g., Petrol_Consumption, Income, Tax, Population_Density, Distance_to_Work).

Tool Usage:

- **WEKA:** Data preprocessing, model training, and performance evaluation.
- **EViews:** Model estimation, diagnostics, and residual analysis.
- **MATLAB:** Regression Learner for model training and validation.

Interpretation: Analyse coefficients, residuals, and fit metrics.

Outcome: Gain insights into the practical use of statistical tools and the application of multiple linear regression for real-world data analysis.

Dataset Description

Dataset Name: Petrol Consumption Dataset

Purpose: To analyse the relationship between multiple independent factors and petrol consumption across regions.

Variables in the Dataset:

1. Dependent Variable:

- **Petrol_Consumption (Y):** The amount of petrol consumed (in litres per capita).

2. Independent Variables:

- **Petrol_Tax:** Tax on petrol (monetary units per litre).
- **Average_Income:** Average annual income per region (monetary units).
- **Paved_Highways:** Total length of paved highways (kilometres).
- **Population_Driver_License:** Percentage of the population holding a driver's license (%).

Key Attributes:

- Data type: Numerical
- Number of observations: 48

Application: Suitable for *multiple linear regression analysis* to determine influential factors on petrol consumption.

! Important Note !

If You Missed the Live Lecture:

A detailed demonstration of the applications (WEKA, EViews, and MATLAB) using the *Petrol Consumption dataset* is available in the video recording.

Please check the video recording on **Blackboard** for step-by-step guidance on:

- Dataset setup
- Regression analysis in each app
- Interpreting the results
- Comparing the tools

Recommendation:

Review the video to ensure you understand the practical application of regression analysis across the tools.

WEKA Overview



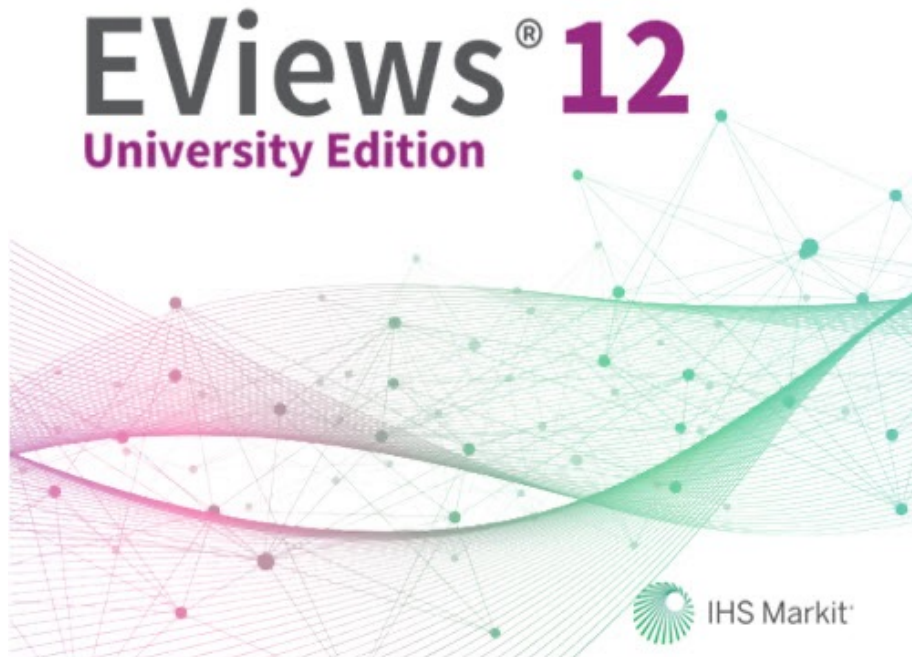
WEKA is a collection of machine learning algorithms for data mining tasks.

Features: Includes tools for data pre-processing, classification, regression, clustering, association rules, and visualisation.

Challenges: Managing large datasets and integration with other programming languages.

Official website: [WEKA Official Site](https://www.cs.waikato.ac.nz/~ml/weka/)

EViews Overview



EViews blends data handling, statistical analysis, and econometric tools.

Features: User-friendly interface, extensive econometric and statistical capabilities, graphing and reporting.

Challenges: Requires foundational knowledge in econometrics, which is complex for beginners.

Official website: [EViews Official Site](https://www.eviews.com/)

MATLAB Overview



MATLAB specialises in matrix operations, numerical analysis, and algorithm development.

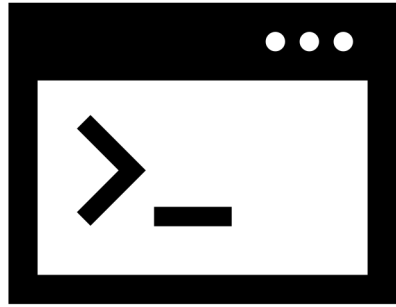
Features: Integrated development environment, rich function library, extensive toolboxes for various applications.

Challenges: High complexity, requires programming skills.

Official website: [MATLAB Official Site](https://www.mathworks.com/)

Comparison of Statistical Analysis Applications

Feature	WEKA	MATLAB	EViews
Primary Focus	Machine learning and data mining	Numerical computing and algorithm design	Econometric and statistical analysis
User Interface	Graphical interface with workflow-oriented tools	Integrated development environment (IDE)	User-friendly with econometrics focus
Statistical Techniques	Supports classification, regression, clustering, and association rule mining	Broad support for statistical analysis, optimization, and visualization	Specialized in econometric models, time series, and forecasting
Data Handling	Handles small to medium datasets efficiently	Handles large datasets; requires sufficient memory	Handles time series and panel data effectively
Ease of Use	Easy for beginners; workflow-based approach	Requires programming skills	Moderate learning curve for statistical tasks
Visualization	Basic visualization; focuses on model performance	Advanced plotting and graphing capabilities	Built-in tools for time series and econometrics
Regression Analysis	Includes linear and multiple regression	Advanced regression capabilities, including nonlinear regression	Robust regression tools for econometric analysis
Tool Availability	Free and open-source	Proprietary; license required	Proprietary; license required
Learning Curve	Low to medium; intuitive interface	Steep; requires programming knowledge	Medium; familiarity with econometrics recommended
Integration	Limited integration with other tools	Strong integration with various programming environments	Integration with standard econometrics software
Output Reports	Basic reports; focused on performance metrics	Customizable reports; export capabilities	Extensive pre-formatted reports and summaries



Programming Tools for Statistical Analysis

Python and R

Python: Versatile Programming for Diverse Applications



Python is a high-level, interpreted programming language known for its simplicity and readability.

Features: Extensive standard library, a large ecosystem of third-party packages, strong in data manipulation, analysis, and machine learning (NumPy, Pandas, Scikit-learn, TensorFlow).

Challenges: Performance limitations with large datasets managing different environments.

Official website: [Python Official Site](https://www.python.org/)

R: The Statistical Programming Language



R is a language and environment for statistical computing and graphics.

Features: Highly extensible, offers a wide variety of statistical and graphical techniques (linear and nonlinear modelling, statistical tests, time series analysis, classification, clustering, etc.).

Challenges: Steep learning curve for beginners, memory management.

Official website: [R Project Official Site](https://www.r-project.org/)